Yufeng (Alex) Xia

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Profile

- MSc by Research student at the Institute for Computing Systems Architecture (ICSA), University of Edinburgh, supervised by Prof. Mahesh K. Marina. My interest lies at the intersection of **wireless systems, systems for AI, and large-scale emulation**. My work spans **AI-RAN**, where live radio processing and data-driven workloads share the same infrastructure under real-time constraints, as well as **emulation and digital twin systems**, the platforms on which RAN and edge-AI algorithms are developed and validated at scale. My research has resulted in publications at top-tier venues including ACM MobiCom.

Education

- **University of Edinburgh — MSc by Research, Computer Science** — Edinburgh, UK
  Supervised by Prof. Mahesh K. Marina. Thesis: Flux-DT: An Elastic, Preemption-Tolerant Digital Twin Runtime for AI-RAN — *Sep 2025 – Sep 2026 (expected)*
- **University of Edinburgh — BSc Computer Science** — Edinburgh, UK
  *Grade: **First Class Honours*** — *Sep 2021 – May 2025*
  - **Relevant courses:** Machine Learning Systems, Machine Learning Practical

Technical Skills

- **Languages** C, C++, Python, Java, Bash.
- **ML & GPU Systems** PyTorch, CUDA programming, GPU profiling and roofline analysis, distributed training at scale, resource scheduling and allocation policies.
- **Performance & Evaluation** Benchmarking and microbenchmarking, system bottleneck analysis, large-scale system evaluation.
- **Infrastructure** Linux, Docker, Kubernetes, Git, CMake, Microsoft Azure, CloudLab.

Research and Project Experience

- **Flux-DT — Elastic, Preemption-Tolerant Digital Twin Runtime for AI-RAN** — **MSc by Research Thesis**
  University of Edinburgh — sole author — *Sep 2025 – Present*
  - **Sharing RAN compute with a stateful workload**: Built a runtime that lets a radio digital twin run on bursty spare compute in the radio path , holding its scene state across interruption without violating live RAN priorities. Prior compute-sharing designs assume the co-located workload is disposable.
  - **Near-real-time control policy**: Designed and implemented the control loop that answers radio-driven resource pressure on the timescale the radio imposes.
  - **Multi-cell fidelity placement algorithm**: Formulated fidelity placement as a joint objective across co-channel cells and **designed a greedy heuristic** deciding which tiles are worth solving under interference
- **EmuRAN — Scalable, Cost-Effective Cloud-based RAN Emulation System** — MobiCom 2026
  *Research Assistant; **owned the end-to-end evaluation*** — *Conditionally Accepted*
  - **Public cloud operation**: Deployed and operated the emulator on public cloud infrastructure, emulating over 10,000 mobile devices, two orders of magnitude larger in scale than existing software solutions.
  - **Fidelity and scalability validation**: Designed and ran the evaluation campaign, covering both fidelity against a real RAN and scaling behaviour under constrained compute.
- **Weaver — Foundation Model Training over AI-RAN Compute Infrastructure** — Under Submission
  Research Assistant; spare-compute controller evaluation and large-scale experiments — *2024 – 2026*

- **RAN-centric spare compute control**: Evaluated the controller deciding how much accelerator capacity an AI workload may harvest from shared RAN infrastructure, a near-real-time resource-allocation problem on GPU-accelerated hardware. On a multi-site testbed it harvests up to **83%** of spare compute without degrading live radio performance.
- **PHY experiments and large-scale training runs**: Ran PHY-layer experiments and GPU profiling to characterise the interference between training workloads and radio processing, and carried out the large-scale foundation-model training runs used in the evaluation.

- **CoSenseAQ — LLVM Auto-Quantization from Sensor Specifications** Under Submission
  Research Assistant, ICSA; first author *Apr 2024 – May 2025*
  - **Sensor specifications as compiler input**: Built an LLVM auto-quantization framework that drives optimisation from sensor specifications, information a compiler cannot recover from source alone, reducing code size and execution time on embedded targets.
  - **Embedded evaluation**: Evaluated on embedded ARM devices, measuring speedup, code size reduction and lower power consumption.

- **Morphling — Emulator for Distributed Machine Learning at the Edge** EdgeSys 2026
  Research Assistant; ***Best Paper Award*** *Oct 2025 – Mar 2026*
  - **Measurement-driven edge emulation**: Contributed to a measurement-driven emulator for distributed ML training on heterogeneous edge devices, supporting up to 1,800 emulated devices on a single GPU server with low latency and thermal error.

Industry Experience

- **Senior Tech — Back-end Development Intern** Suzhou, China
  Backend and QA work on Uniqlo's logistics and mobile products *Jun 2023 – Sep 2023*
  - **Express delivery platform**: Contributed to the backend architecture upgrade of Uniqlo's express delivery platform, migrating a live production system from JFinal to Spring Boot and rewriting its interface layer; took part in code review and release testing alongside the permanent team. Java, Spring Boot, MySQL, Redis, Kafka
  - **Android application testing**: Tested the Uniqlo Android app for performance, reliability and user experience, identifying and reporting defects ahead of release.

Selected Publications

EmuRAN: Scalable and Cost-Effective Cloud-based RAN Emulation System *MobiCom'26*
Ujjwal Pawar, Andrew E. Ferguson, ***Yufeng Xia****, Xenofon Foukas, Mahesh Marina, Bozidar Radunovic*

Morphling: Emulator for Distributed Machine Learning at the Edge *EdgeSys'26 (Best Paper)*
Leyang Xue, ***Yufeng Xia****, Ismaeel Bashir, Eren Mendi, Jiaxun Yang, Myungjin Lee, Mahesh Marina*

Wasp: Foundation Model Training System for the Edge *SEC'26*
Leyang Xue, ***Yufeng Xia****, Myungjin Lee, Mahesh Marina*

CoSenseAQ: Compiler Optimizations using Sensor Technical Specifications with Automatic Quantization *under submission*
Yufeng Xia*, Pei Mu, Antonio Barbalace*

Awards, Languages and Service

- **Awards Best Paper Award**, EdgeSys 2026 (Workshop co-located with ACM MobiSys 2026), for *Morphling.* Student Travel Grant, ACM MobiSys 2026.
- **Languages** Mandarin (native), English (fluent). IELTS 7.0; GRE 336.
- **Service** Artifact Evaluation Committee Member, ACM CGO 2027. Student volunteer, EuroSys 2026 (on-site conference operations).

References

- **Prof. Mahesh K. Marina** University of Edinburgh, MSc by Research supervisor · mahesh@ed.ac.uk
- **Dr Antonio Barbalace** University of Edinburgh, BSc and RA supervisor · antonio.barbalace@ed.ac.uk